

MANGNAI, China

WMO: 518860

Lat: 38.250N

Lon: 90.850E

Elev: 2945

StdP: 70.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-17.7	-34.2	0.2	-7.5	-31.6	0.3	-9.2	12.6	-3.3	9.0	-2.9	0.7	140	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	26.5	9.8	24.9	9.2	23.5	8.6	12.1	20.8	10.8	20.5	9.9	20.0	3.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.6	10.0	15.3	7.0	8.9	14.3	5.5	8.1	14.0	43.9	20.9	40.0	20.5	37.4	20.0	16.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.1	6.5	DB	-21.8	29.5	1.5	1.6	-22.9	30.6	-23.7	31.6	-24.6	32.5	-25.7	33.6	(4)
(5)			WB	-22.7	13.1	1.4	1.7	-23.7	14.3	-24.4	15.2	-25.2	16.2	-26.2	17.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-9.9	-5.3	0.3	6.1	10.8	14.9	17.5	16.8	11.9	4.5	-3.0	-9.0	(6)	
	DBStd	9.99	3.22	3.59	3.56	3.60	2.88	2.50	2.66	2.39	2.55	3.06	3.44	2.90	(7)	
	HDD10.0	2653	617	428	299	123	26	1	0	0	9	172	389	588	(8)	
	HDD18.3	5017	876	662	558	366	234	106	49	59	193	429	639	847	(9)	
	CDD10.0	713	0	0	0	8	50	147	231	211	66	1	0	0	(10)	
	CDD18.3	36	0	0	0	0	0	2	22	12	1	0	0	0	(11)	
	CDH23.3	350	0	0	0	1	3	27	192	124	3	0	0	0	(12)	
	CDH26.7	44	0	0	0	0	0	1	28	15	0	0	0	0	(13)	
(14) Wind	WSAvg	2.4	1.5	2.1	2.8	3.2	3.0	3.0	2.8	2.7	2.5	2.1	1.7	1.5	(14)	
(15) Precipitation	PrecAvg	49	1	1	1	2	5	16	14	6	2	1	0	0	(15)	
	PrecMax	106	5	3	4	15	35	54	31	26	12	5	2	2	(16)	
	PrecMin	19	0	0	0	0	0	2	2	0	0	0	0	0	(17)	
	PrecStd	20	1	1	1	3	7	13	8	5	3	1	0	0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	8.2	15.3	20.5	23.1	25.6	29.4	28.5	23.6	17.4	9.7	2.5	(19)	
		MCWB	-4.6	-1.3	2.3	4.8	6.3	8.5	12.0	10.9	8.4	4.1	-0.1	-4.1	(20)	
	2%	DB	-0.2	5.4	12.1	18.0	21.1	24.0	27.1	26.2	21.6	15.1	7.4	0.1	(21)	
		MCWB	-6.1	-2.8	0.5	3.8	5.6	8.0	10.2	9.5	7.3	2.9	-1.4	-5.6	(22)	
	5%	DB	-2.1	3.4	10.0	16.0	19.4	22.6	25.5	24.5	19.8	13.3	5.6	-1.8	(23)	
		MCWB	-7.3	-4.0	-0.5	2.9	5.0	7.6	9.8	9.0	6.3	2.1	-2.4	-6.9	(24)	
	10%	DB	-3.9	1.6	8.0	14.1	17.8	21.1	23.8	23.1	18.3	11.5	3.9	-3.3	(25)	
		MCWB	-8.5	-5.0	-1.5	2.0	4.4	7.0	9.1	8.6	5.6	1.2	-3.4	-7.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.4	-1.2	2.5	5.2	7.3	10.7	14.2	13.4	10.8	4.7	0.1	-3.8	(27)	
		MCDB	1.9	7.6	13.8	19.0	19.5	19.4	24.1	23.2	18.5	13.6	9.1	1.9	(28)	
	2%	WB	-5.9	-2.7	0.9	4.2	6.2	9.5	12.6	11.7	8.6	3.5	-1.3	-5.5	(29)	
		MCDB	-0.5	4.9	11.3	16.8	18.5	18.3	22.3	20.1	18.3	13.4	6.8	-0.3	(30)	
	5%	WB	-7.2	-3.8	-0.3	3.2	5.5	8.7	11.2	10.6	7.5	2.5	-2.3	-6.8	(31)	
		MCDB	-2.3	3.2	9.6	14.9	17.3	18.0	21.1	20.2	17.0	12.5	5.3	-2.0	(32)	
	10%	WB	-8.4	-4.9	-1.3	2.3	4.9	8.1	10.3	9.7	6.6	1.6	-3.3	-7.8	(33)	
		MCDB	-4.1	1.4	7.7	13.3	16.2	18.1	20.5	20.2	16.0	11.0	3.7	-3.5	(34)	
(35) Mean Daily Temperature Range	MDBR		12.1	12.9	13.8	13.7	12.3	11.1	11.1	11.8	12.6	13.7	13.0	11.8	(35)	
	5% DB	MCDBR	13.3	14.7	16.4	16.0	14.7	13.6	13.6	14.0	14.7	15.2	14.4	12.6	(36)	
		MCWBR	9.2	9.1	9.0	7.8	6.4	5.1	4.8	5.2	6.3	7.6	8.6	8.7	(37)	
	5% WB	MCDBR	13.0	14.3	15.5	14.7	12.8	10.4	11.2	11.0	11.5	14.2	14.1	12.2	(38)	
		MCWBR	9.0	8.9	8.6	7.1	5.7	4.1	4.1	4.3	5.0	7.1	8.5	8.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.228	0.269	0.324	0.389	0.413	0.427	0.415	0.396	0.353	0.288	0.246	0.223	(40)	
	taud		2.444	2.315	2.112	1.924	1.925	1.956	2.053	2.095	2.182	2.313	2.413	2.461	(41)	
	Ebn at Noon		982	965	931	885	868	854	861	869	889	927	946	965	(42)	
	Edh at Noon		80	105	144	186	189	184	165	155	133	104	82	73	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.99	3.94	5.31	6.37	6.98	6.82	6.57	6.32	5.57	4.63	3.41	2.74	(44)	
	RadStd		0.16	0.16	0.17	0.32	0.30	0.43	0.44	0.25	0.25	0.17	0.14	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+2.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page